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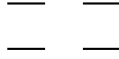
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1 Orbitals And Their Symmetry

We use the following order of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_2^* & \text{---} & \text{---} & a_2^* \\ a_2 & \text{---} & \text{---} & b_2 \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:



where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital order within the group of monomer orbitals follows the above mentioned definition. The transformation of

monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$a_2^{*l} = +b_1^* + a_2^* \tag{1a}$$

$$b_2^{*l} = +a_1^* + b_2^* \tag{1b}$$

$$a_2^l = +b_1 + a_2 \tag{1c}$$

$$b_2^l = +a_1 + b_2 \tag{1d}$$

$$a_2^{*r} = -b_1^* + a_2^* \tag{1e}$$

$$b_2^{*r} = -a_1^* + b_2^* \tag{1f}$$

$$a_2^r = -b_1 + a_2 \tag{1g}$$

$$b_2^r = -a_1 + b_2 \tag{1h}$$

2 Linear Combinations: 2 Monomer- / 1 Dimer-States

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^{*l}) \cdot (a_2^l) \cdot (b_2^l)}_{l: a_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*l})| \\
& = + |(a_1 + b_2) \cdot (+a_1^* + b_2^*) \cdot (+b_1 + a_2) \cdot (+b_1^* + a_2^*)| \\
& = + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& \quad + \left| \underbrace{a_1 b_2^* b_1 b_1^*}_{b_2} \right| + \left| \underbrace{a_1 b_2^* b_1 a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& \quad + \left| \underbrace{b_2 a_1^* b_1 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 a_1^* b_1 a_2^*}_{a_1} \right| + \left| \underbrace{b_2 a_1^* a_2 b_1^*}_{a_1} \right| + \left| \underbrace{b_2 a_1^* a_2 a_2^*}_{b_2} \right| \\
& \quad + \left| \underbrace{b_2 b_2^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{b_2 b_2^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\
& = + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& \quad + \left| \underbrace{a_1 b_1^* b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& \quad - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\
& \quad + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(b_2^{*r}) \cdot (a_2^{*r}) \cdot (a_2^r) \cdot (b_2^r)}_{r: a_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*r})| \\
& = + |(-a_1 + b_2) \cdot (-a_1^* + b_2^*) \cdot (-b_1 + a_2) \cdot (-b_1^* + a_2^*)|
\end{aligned}$$

$$\begin{aligned}
&= + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
&\quad - \left| \underbrace{a_1 b_2^* b_1 b_1^*}_{b_2} \right| + \left| \underbrace{a_1 b_2^* b_1 a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
&\quad - \left| \underbrace{b_2 a_1^* b_1 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 a_1^* b_1 a_2^*}_{a_1} \right| + \left| \underbrace{b_2 a_1^* a_2 b_1^*}_{a_1} \right| - \left| \underbrace{b_2 a_1^* a_2 a_2^*}_{b_2} \right| \\
&\quad + \left| \underbrace{b_2 b_2^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{b_2 b_2^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

$$\begin{aligned}
&= + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
&\quad - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
&\quad + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\
&\quad + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{\text{l: } a_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{\text{r: } a_1} \Rightarrow + |(a_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(+b_1 + a_2) \cdot (-b_1 + a_2) \cdot (+b_1^* + a_2^*) \cdot (-b_1^* + a_2^*)| \\
& = + \left| \underbrace{b_1 a_2 b_1^* a_2^*}_{a_1} \right| - \left| \underbrace{b_1 a_2 a_2^* b_1^*}_{a_1} \right| - \left| \underbrace{a_2 b_1 b_1^* a_2^*}_{a_1} \right| + \left| \underbrace{a_2 b_1 a_2^* b_1^*}_{a_1} \right| \\
& = - \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{l: a_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{r: b_1} \Rightarrow + |(b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*l})| \\
& = + |(-a_1^* + b_2^*) \cdot (+b_1 + a_2) \cdot (-b_1 + a_2) \cdot (+b_1^* + a_2^*)| \\
& = - \left| \underbrace{a_1^* b_1 a_2 b_1^*}_{a_2} \right| - \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + \left| \underbrace{a_1^* a_2 b_1 b_1^*}_{a_2} \right| + \left| \underbrace{a_1^* a_2 b_1 a_2^*}_{b_1} \right| \\
& \quad + \left| \underbrace{b_2^* b_1 a_2 b_1^*}_{b_1} \right| + \left| \underbrace{b_2^* b_1 a_2 a_2^*}_{a_2} \right| - \left| \underbrace{b_2^* a_2 b_1 b_1^*}_{b_1} \right| - \left| \underbrace{b_2^* a_2 b_1 a_2^*}_{a_2} \right| \\
& = + \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| \\
& \quad - \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| - \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& = +2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\
& \quad + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{l: b_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{r: a_1} \Rightarrow + |(b_2^{*l}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*r})| \\
& = + |(+a_1^* + b_2^*) \cdot (+b_1 + a_2) \cdot (-b_1 + a_2) \cdot (-b_1^* + a_2^*)| \\
& = - \left| \underbrace{a_1^* b_1 a_2 b_1^*}_{a_2} \right| + \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + \left| \underbrace{a_1^* a_2 b_1 b_1^*}_{a_2} \right| - \left| \underbrace{a_1^* a_2 b_1 a_2^*}_{b_1} \right| \\
& \quad - \left| \underbrace{b_2^* b_1 a_2 b_1^*}_{b_1} \right| + \left| \underbrace{b_2^* b_1 a_2 a_2^*}_{a_2} \right| + \left| \underbrace{b_2^* a_2 b_1 b_1^*}_{b_1} \right| - \left| \underbrace{b_2^* a_2 b_1 a_2^*}_{a_2} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| \\
& + \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| + \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{l: a_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{r: b_1} \Rightarrow + |(b_2^r) \cdot (a_2^l) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(-a_1 + b_2) \cdot (+b_1 + a_2) \cdot (+b_1^* + a_2^*) \cdot (-b_1^* + a_2^*)| \\
\\
& = - \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + \left| \underbrace{a_1 b_1 a_2^* b_1^*}_{a_2} \right| - \left| \underbrace{a_1 a_2 b_1^* a_2^*}_{b_1} \right| + \left| \underbrace{a_1 a_2 a_2^* b_1^*}_{b_1} \right| \\
& \quad + \left| \underbrace{b_2 b_1 b_1^* a_2^*}_{b_1} \right| - \left| \underbrace{b_2 b_1 a_2^* b_1^*}_{b_1} \right| + \left| \underbrace{b_2 a_2 b_1^* a_2^*}_{a_2} \right| - \left| \underbrace{b_2 a_2 a_2^* b_1^*}_{a_2} \right| \\
\\
& = - \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| \\
& \quad + \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| + \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& = -2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\
\\
& + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{l: b_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{r: a_1} \Rightarrow + |(b_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(+a_1 + b_2) \cdot (-b_1 + a_2) \cdot (+b_1^* + a_2^*) \cdot (-b_1^* + a_2^*)| \\
\\
& = - \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + \left| \underbrace{a_1 b_1 a_2^* b_1^*}_{a_2} \right| + \left| \underbrace{a_1 a_2 b_1^* a_2^*}_{b_1} \right| - \left| \underbrace{a_1 a_2 a_2^* b_1^*}_{b_1} \right| \\
& \quad - \left| \underbrace{b_2 b_1 b_1^* a_2^*}_{b_1} \right| + \left| \underbrace{b_2 b_1 a_2^* b_1^*}_{b_1} \right| + \left| \underbrace{b_2 a_2 b_1^* a_2^*}_{a_2} \right| - \left| \underbrace{b_2 a_2 a_2^* b_1^*}_{a_2} \right|
\end{aligned}$$

$$\begin{aligned}
= & - \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| \\
& - \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| + \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (a_2^l)}_{l: a_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{r: a_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^{*l})| \\
& = + |(-a_1 + b_2) \cdot (-a_1^* + b_2^*) \cdot (+b_1 + a_2) \cdot (+b_1^* + a_2^*)| \\
& = + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& \quad - \left| \underbrace{a_1 b_2^* b_1 b_1^*}_{b_2} \right| - \left| \underbrace{a_1 b_2^* b_1 a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& \quad - \left| \underbrace{b_2 a_1^* b_1 b_1^*}_{b_2} \right| - \left| \underbrace{b_2 a_1^* b_1 a_2^*}_{a_1} \right| - \left| \underbrace{b_2 a_1^* a_2 b_1^*}_{a_1} \right| - \left| \underbrace{b_2 a_1^* a_2 a_2^*}_{b_2} \right| \\
& \quad + \left| \underbrace{b_2 b_2^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{b_2 b_2^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\
& = + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& \quad - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& \quad + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\
& \quad + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} \\
& + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{l: a_1} \cdot \underbrace{(a_2^{*r}) \cdot (a_2^r)}_{r: a_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^r) \cdot (a_2^{*r})| \\
& = + |(+a_1 + b_2) \cdot (+a_1^* + b_2^*) \cdot (-b_1 + a_2) \cdot (-b_1^* + a_2^*)|
\end{aligned}$$

$$\begin{aligned}
&= + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
&\quad + \left| \underbrace{a_1 b_2^* b_1 b_1^*}_{b_2} \right| - \left| \underbrace{a_1 b_2^* b_1 a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
&\quad + \left| \underbrace{b_2 a_1^* b_1 b_1^*}_{b_2} \right| - \left| \underbrace{b_2 a_1^* b_1 a_2^*}_{a_1} \right| - \left| \underbrace{b_2 a_1^* a_2 b_1^*}_{a_1} \right| + \left| \underbrace{b_2 a_1^* a_2 a_2^*}_{b_2} \right| \\
&\quad + \left| \underbrace{b_2 b_2^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{b_2 b_2^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

$$\begin{aligned}
&= + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
&\quad + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
&\quad - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\
&\quad + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \hline \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{\text{l: } b_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r)| \\
& = + |(+a_1^* + b_2^*) \cdot (-a_1^* + b_2^*) \cdot (+b_1 + a_2) \cdot (-b_1 + a_2)| \\
& = + \left| \underbrace{a_1^* b_2^* b_1 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2^* a_2 b_1}_{a_1} \right| - \left| \underbrace{b_2^* a_1^* b_1 a_2}_{a_1} \right| + \left| \underbrace{b_2^* a_1^* a_2 b_1}_{a_1} \right| \\
& = - \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{l: b_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{r: b_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*r})| \\
& = + |(-a_1 + b_2) \cdot (+a_1^* + b_2^*) \cdot (+b_1 + a_2) \cdot (-b_1^* + a_2^*)| \\
& = + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} - \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} + \underbrace{a_1 a_1^* a_2 b_1^*}_{b_2} - \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& \quad + \left| \underbrace{a_1 b_2^* b_1 b_1^*}_{b_2} - \underbrace{a_1 b_2^* b_1 a_2^*}_{a_1} + \underbrace{a_1 b_2^* a_2 b_1^*}_{a_1} - \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& \quad - \left| \underbrace{b_2 a_1^* b_1 b_1^*}_{b_2} + \underbrace{b_2 a_1^* b_1 a_2^*}_{a_1} - \underbrace{b_2 a_1^* a_2 b_1^*}_{a_1} + \underbrace{b_2 a_1^* a_2 a_2^*}_{b_2} \right| \\
& \quad - \left| \underbrace{b_2 b_2^* b_1 b_1^*}_{a_1} + \underbrace{b_2 b_2^* b_1 a_2^*}_{b_2} - \underbrace{b_2 b_2^* a_2 b_1^*}_{b_2} + \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\
& = + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} - \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} - \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} - \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& \quad + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} + \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} + \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} - \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& \quad + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} + \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} + \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} - \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\
& \quad - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} + \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} + \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} + \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{l: b_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{r: b_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*l})| \\
& = + |(a_1 + b_2) \cdot (-a_1^* + b_2^*) \cdot (-b_1 + a_2) \cdot (+b_1^* + a_2^*)|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 b_1^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& - \left| \underbrace{a_1 b_2^* b_1 b_1^*}_{b_2} \right| - \left| \underbrace{a_1 b_2^* b_1 a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& + \left| \underbrace{b_2 a_1^* b_1 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 a_1^* b_1 a_2^*}_{a_1} \right| - \left| \underbrace{b_2 a_1^* a_2 b_1^*}_{a_1} \right| - \left| \underbrace{b_2 a_1^* a_2 a_2^*}_{b_2} \right| \\
& - \left| \underbrace{b_2 b_2^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{b_2 b_2^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 b_1^*}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\
& - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
& - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\
& - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

(irreducible)

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \hline \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (a_2^l)}_{l: b_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{r: a_1} \Rightarrow + |(b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l)| \\
& = + |(-a_1 + b_2) \cdot (+a_1^* + b_2^*) \cdot (-a_1^* + b_2^*) \cdot (+b_1 + a_2)| \\
\\
& = - \left| \underbrace{a_1 a_1^* b_2^* b_1}_{a_2} \right| - \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + \left| \underbrace{a_1 b_2^* a_1^* b_1}_{a_2} \right| + \left| \underbrace{a_1 b_2^* a_1^* a_2}_{b_1} \right| \\
& \quad + \left| \underbrace{b_2 a_1^* b_2^* b_1}_{b_1} \right| + \left| \underbrace{b_2 a_1^* b_2^* a_2}_{a_2} \right| - \left| \underbrace{b_2 b_2^* a_1^* b_1}_{b_1} \right| - \left| \underbrace{b_2 b_2^* a_1^* a_2}_{a_2} \right| \\
\\
& = + \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| \\
& \quad - \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| - \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \hline \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{l: a_1} \cdot \underbrace{(b_2^{*r}) \cdot (a_2^r)}_{r: b_1} \Rightarrow + |(b_2^l) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^r)| \\
& = + |(a_1 + b_2) \cdot (+a_1^* + b_2^*) \cdot (-a_1^* + b_2^*) \cdot (-b_1 + a_2)| \\
\\
& = - \left| \underbrace{a_1 a_1^* b_2^* b_1}_{a_2} \right| + \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + \left| \underbrace{a_1 b_2^* a_1^* b_1}_{a_2} \right| - \left| \underbrace{a_1 b_2^* a_1^* a_2}_{b_1} \right| \\
& \quad - \left| \underbrace{b_2 a_1^* b_2^* b_1}_{b_1} \right| + \left| \underbrace{b_2 a_1^* b_2^* a_2}_{a_2} \right| + \left| \underbrace{b_2 b_2^* a_1^* b_1}_{b_1} \right| - \left| \underbrace{b_2 b_2^* a_1^* a_2}_{a_2} \right|
\end{aligned}$$

$$\begin{aligned}
= & + \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| \\
& + \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| + \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right|
\end{aligned}$$

summarized:

$$= +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{\text{l: } b_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{\text{r: } b_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(+a_1 + b_2) \cdot (-a_1 + b_2) \cdot (+b_1^* + a_2^*) \cdot (-b_1^* + a_2^*)| \\
& = + \left| \underbrace{a_1 b_2 b_1^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_2 a_2^* b_1^*}_{a_1} \right| - \left| \underbrace{b_2 a_1 b_1^* a_2^*}_{a_1} \right| + \left| \underbrace{b_2 a_1 a_2^* b_1^*}_{a_1} \right| \\
& = - \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|$$

$$\begin{aligned}
& + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(a_2^{*l}) \cdot (b_2^l)}_{l: b_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{r: a_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (b_2^{*r}) \cdot (a_2^{*l})| \\
& = + |(a_1 + b_2) \cdot (-a_1 + b_2) \cdot (-a_1^* + b_2^*) \cdot (+b_1^* + a_2^*)| \\
& = - \left| \underbrace{a_1 b_2 a_1^* b_1^*}_{a_2} \right| - \left| \underbrace{a_1 b_2 a_1^* a_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_2 b_2^* b_1^*}_{b_1} \right| + \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
& \quad + \left| \underbrace{b_2 a_1 a_1^* b_1^*}_{a_2} \right| + \left| \underbrace{b_2 a_1 a_1^* a_2^*}_{b_1} \right| - \left| \underbrace{b_2 a_1 b_2^* b_1^*}_{b_1} \right| - \left| \underbrace{b_2 a_1 b_2^* a_2^*}_{a_2} \right| \\
& = - \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
& \quad - \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right|
\end{aligned}$$

summarized:

$$\begin{aligned}
& = -2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
& \quad + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{l: a_1} \cdot \underbrace{(a_2^{*r}) \cdot (b_2^r)}_{r: b_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (a_2^{*r})| \\
& = + |(a_1 + b_2) \cdot (-a_1 + b_2) \cdot (+a_1^* + b_2^*) \cdot (-b_1^* + a_2^*)| \\
& = - \left| \underbrace{a_1 b_2 a_1^* b_1^*}_{a_2} \right| + \left| \underbrace{a_1 b_2 a_1^* a_2^*}_{b_1} \right| - \left| \underbrace{a_1 b_2 b_2^* b_1^*}_{b_1} \right| + \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
& \quad + \left| \underbrace{b_2 a_1 a_1^* b_1^*}_{a_2} \right| - \left| \underbrace{b_2 a_1 a_1^* a_2^*}_{b_1} \right| + \left| \underbrace{b_2 a_1 b_2^* b_1^*}_{b_1} \right| - \left| \underbrace{b_2 a_1 b_2^* a_2^*}_{a_2} \right|
\end{aligned}$$

$$\begin{aligned}
&= - \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
&\quad - \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right|
\end{aligned}$$

summarized:

$$= -2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right|$$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \underbrace{(b_2^{*l}) \cdot (b_2^l)}_{\text{l: } a_1} \cdot \underbrace{(b_2^{*r}) \cdot (b_2^r)}_{\text{r: } a_1} \Rightarrow + |(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r})| \\
& = + |(+a_1 + b_2) \cdot (-a_1 + b_2) \cdot (+a_1^* + b_2^*) \cdot (-a_1^* + b_2^*)| \\
& = + \left| \underbrace{a_1 b_2 a_1^* b_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_2 b_2^* a_1^*}_{a_1} \right| - \left| \underbrace{b_2 a_1 a_1^* b_2^*}_{a_1} \right| + \left| \underbrace{b_2 a_1 b_2^* a_1^*}_{a_1} \right| \\
& = - \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right|
\end{aligned}$$

summarized:

$$= -4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right|$$

3 Linear Combinations: 4 Monomer- / 2 Dimer-States

$$+ \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1

$$= + (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*l})|) + (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= +2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right|$$

$$+ 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|$$

$$+ \begin{pmatrix} \uparrow & \uparrow & - & - \\ \uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & - & \uparrow & \uparrow \\ \uparrow\uparrow & \uparrow\uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1

$$= + (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*l})|) - (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= +2 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + 2 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + 2 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right|$$

$$- 2 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - 2 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| + 2 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right|$$

$$+ \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow \uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = a_1

$$= \qquad \qquad \qquad + (|(a_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|) \qquad \qquad + \qquad \qquad (|(a_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad -8 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*l})|) \qquad \qquad + \qquad \qquad (|(b_2^{*l}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad + 4 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| \qquad \qquad - \qquad \qquad 4 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right|$$

$$+ \begin{pmatrix} \text{---} & \uparrow & \uparrow & \text{---} \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{---} & \text{---} & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*l})|) \qquad \qquad - \qquad \qquad (|(b_2^{*l}) \cdot (a_2^l) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad - 4 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| \qquad \qquad - \qquad \qquad 4 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right|$$

$$+ \begin{pmatrix} \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= + (|(b_2^r) \cdot (a_2^l) \cdot (a_2^{*l}) \cdot (a_2^{*r})|) + (|(b_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|)$$

$$= -4 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 4 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|$$

$$+ \begin{pmatrix} \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} & \overline{\uparrow} \\ \uparrow & \uparrow & \uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= + (|(b_2^r) \cdot (a_2^l) \cdot (a_2^{*l}) \cdot (a_2^{*r})|) - (|(b_2^l) \cdot (a_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r})|)$$

$$= +4 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 4 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right|$$

$$+ \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1

$$= + (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^{*l})|) + (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= +2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right|$$

$$- 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|$$

$$+ \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1

$$= + (|(b_2^r) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^{*l})|) - (|(b_2^l) \cdot (b_2^{*l}) \cdot (a_2^r) \cdot (a_2^{*r})|)$$

$$= +2 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - 2 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right|$$

$$+ 2 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + 2 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| + 2 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right|$$

$$+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = b_1

$$= \qquad \qquad \qquad + (|(b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r)|) \qquad \qquad + \qquad \qquad (|(b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l) \cdot (a_2^r)|)$$

$$= \qquad \qquad \qquad -8 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right|$$

$$+ \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_1

$$= + (|(b_2^r) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*r})|) + (|(b_2^l) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*l})|)$$

$$= +2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right|$$

$$+ 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right|$$

$$+ \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = b_1

$$= + (|(b_2^r) \cdot (b_2^{*l}) \cdot (a_2^l) \cdot (a_2^{*r})|) - (|(b_2^l) \cdot (b_2^{*r}) \cdot (a_2^r) \cdot (a_2^{*l})|)$$

$$= -2 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + 2 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right|$$

$$+ 2 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - 2 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| + 2 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right|$$

$$+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l)|) \qquad \qquad + \qquad \qquad (|(b_2^l) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^r)|)$$

$$= \qquad \qquad \qquad + 4 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| \qquad \qquad - \qquad \qquad 4 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right|$$

$$+ \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{array} \right)$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^l)|) \qquad \qquad - \qquad \qquad (|(b_2^l) \cdot (b_2^{*l}) \cdot (b_2^{*r}) \cdot (a_2^r)|)$$

$$= \qquad \qquad \qquad - 4 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| \qquad \qquad - \qquad \qquad 4 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right|$$

enthaltene Monomer-Symmetrien = b_1

$$= \quad + \left(\left| (b_2^l) \cdot (b_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r}) \right| \right) \quad + \quad \left(\left| (b_2^l) \cdot (b_2^r) \cdot (a_2^{*l}) \cdot (a_2^{*r}) \right| \right)$$

$$= -8 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|$$

$$+ \begin{pmatrix} \overline{} & \uparrow & \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \overline{} & \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*r}) \cdot (a_2^{*l})|) \qquad \qquad + \qquad \qquad (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad -4 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| \qquad \qquad + \qquad \qquad 4 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right|$$

$$+ \begin{pmatrix} \overline{} & \uparrow & \uparrow & \overline{} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \overline{} & \overline{} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

enthaltene Monomer-Symmetrien = a_1, b_1

$$= \qquad \qquad \qquad + (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*r}) \cdot (a_2^{*l})|) \qquad \qquad - \qquad \qquad (|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (a_2^{*r})|)$$

$$= \qquad \qquad \qquad +4 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| \qquad \qquad + \qquad \qquad 4 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right|$$

enthaltene Monomer-Symmetrien = a_1

$$+ \left(|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r})| \right) \quad + \quad \left(|(b_2^l) \cdot (b_2^r) \cdot (b_2^{*l}) \cdot (b_2^{*r})| \right)$$

$$= -8 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right|$$

4 Linear Combinations: 8 Monomer- / 4 Dimer-States

$i^3 a_1 * i^3 a_1$:

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| \\
& \quad + 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| \\
& \quad - 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

$i^3 a_1 * e^3 a_1$:

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ -\uparrow & \uparrow\uparrow & \uparrow\uparrow & -\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ -\uparrow & \uparrow\uparrow & -\uparrow & \uparrow\uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| \right) - \left(\begin{array}{l} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) - \left(+4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| \right)$$

$$= -4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + 2 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| \\ + 2 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - 2 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| - 2 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| \\ + 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + 4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|$$

$$\begin{aligned}
&= +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\
&\quad - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
&\quad - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

$$e^3 a_1 * i^3 a_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ -\uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ -\uparrow & \uparrow\uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ -\uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ -\uparrow & \uparrow\uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - 2 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| \\
& \quad - 2 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| + 2 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + 2 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| + 2 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| \\
& \quad - 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + 4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

$$e^3 a_1 * e^3 a_1:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ -\uparrow & \uparrow\uparrow & \uparrow\uparrow & -\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ -\uparrow & \uparrow\uparrow & -\uparrow & \uparrow\uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| \right) + \left(\begin{array}{l} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) + \left(\begin{array}{l} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) + \left(-4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| \right)$$

$$= -4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|$$

$$\begin{aligned}
&= +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

$$\begin{aligned}
&= +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\
&\quad - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\
&\quad - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

$$e^3 b_1 * e^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| \\
& + 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| \\
& + 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|
\end{aligned}$$

$$e^3 b_1 * i^3 b_1:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| \right) + \left(\begin{array}{l} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) - \left(\begin{array}{l} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) - \left(+4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| \right)$$

$$= -4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - 2 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + 2 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + 2 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| - 2 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + 4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|$$

$$\begin{aligned}
&= +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\
&\quad + 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

$$i^3 b_1 * e^3 b_1:$$

$$\begin{aligned}
& + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \right) + \left(\begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \right) - \left(\begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& - \left(+4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + 2 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| \\
& - 2 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| + 2 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - 2 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| + 2 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| \\
& + 2 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + 4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|
\end{aligned}$$

$$i^3 b_1 * i^3 b_1:$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \left(-4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| \right) \\ - \left(\begin{array}{c} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\ - \left(\begin{array}{c} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\ + \left(-4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| \right)$$

$$= -4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| \\ - 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| \\ - 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|$$

5 Linear Combinations: 16 Monomer- / 8 Dimer-States

identical $i^3 a_1$ and $i^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} \\
& + \left(-4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| \\
& \quad + 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| \\
& \quad - 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

plus combination: i^3a_1 and $e^3a_1 + e^3a_1$ and i^3a_1

$$\begin{aligned}
 &+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) \\
 &+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow \uparrow & \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow \uparrow & \downarrow \uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow \uparrow & \uparrow & \uparrow & \downarrow \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow \uparrow & \uparrow & \downarrow \uparrow \end{array} \right)
 \end{aligned}$$

$$\begin{aligned}
& + \left(-4 \underbrace{|a_1 a_1^* b_2 b_2^*|}_{a_1} \right) \\
& - \left(\underbrace{|a_1 a_1^* b_1 b_1^*|}_{a_1} - \underbrace{|a_1 a_1^* b_1 a_2^*|}_{b_2} + \underbrace{|a_1 a_1^* b_1^* a_2|}_{b_2} - \underbrace{|a_1 a_1^* a_2 a_2^*|}_{a_1} \right. \\
& + \underbrace{|a_1 b_1 b_1^* b_2^*|}_{b_2} - \underbrace{|a_1 b_1 b_2^* a_2^*|}_{a_1} + \underbrace{|a_1 b_1^* b_2^* a_2|}_{a_1} + \underbrace{|a_1 b_2^* a_2 a_2^*|}_{b_2} \\
& - \underbrace{|a_1^* b_1 b_1^* b_2|}_{b_2} + \underbrace{|a_1^* b_1 b_2 a_2^*|}_{a_1} - \underbrace{|a_1^* b_1^* b_2 a_2|}_{a_1} - \underbrace{|a_1^* b_2 a_2 a_2^*|}_{b_2} \\
& - \underbrace{|b_1 b_1^* b_2 b_2^*|}_{a_1} - \underbrace{|b_1 b_2 b_2^* a_2^*|}_{b_2} + \underbrace{|b_1^* b_2 b_2^* a_2|}_{b_2} - \underbrace{|b_2 b_2^* a_2 a_2^*|}_{a_1} \Bigg) \\
& + \left(\underbrace{|a_1 a_1^* b_1 b_1^*|}_{a_1} - \underbrace{|a_1 a_1^* b_1 a_2^*|}_{b_2} + \underbrace{|a_1 a_1^* b_1^* a_2|}_{b_2} + \underbrace{|a_1 a_1^* a_2 a_2^*|}_{a_1} \right. \\
& + \underbrace{|a_1 b_1 b_1^* b_2^*|}_{b_2} + \underbrace{|a_1 b_1 b_2^* a_2^*|}_{a_1} - \underbrace{|a_1 b_1^* b_2^* a_2|}_{a_1} + \underbrace{|a_1 b_2^* a_2 a_2^*|}_{b_2} \\
& - \underbrace{|a_1^* b_1 b_1^* b_2|}_{b_2} - \underbrace{|a_1^* b_1 b_2 a_2^*|}_{a_1} + \underbrace{|a_1^* b_1^* b_2 a_2|}_{a_1} - \underbrace{|a_1^* b_2 a_2 a_2^*|}_{b_2} \\
& + \underbrace{|b_1 b_1^* b_2 b_2^*|}_{a_1} - \underbrace{|b_1 b_2 b_2^* a_2^*|}_{b_2} + \underbrace{|b_1^* b_2 b_2^* a_2|}_{b_2} + \underbrace{|b_2 b_2^* a_2 a_2^*|}_{a_1} \Bigg) \\
& - \left(+4 \underbrace{|b_1 b_1^* a_2 a_2^*|}_{a_1} \right) \\
& + \left(-4 \underbrace{|a_1 a_1^* b_2 b_2^*|}_{a_1} \right) \\
& + \left(\underbrace{|a_1 a_1^* b_1 b_1^*|}_{a_1} + \underbrace{|a_1 a_1^* b_1 a_2^*|}_{b_2} - \underbrace{|a_1 a_1^* b_1^* a_2|}_{b_2} + \underbrace{|a_1 a_1^* a_2 a_2^*|}_{a_1} \right. \\
& - \underbrace{|a_1 b_1 b_1^* b_2^*|}_{b_2} + \underbrace{|a_1 b_1 b_2^* a_2^*|}_{a_1} - \underbrace{|a_1 b_1^* b_2^* a_2|}_{a_1} - \underbrace{|a_1 b_2^* a_2 a_2^*|}_{b_2} \\
& + \underbrace{|a_1^* b_1 b_1^* b_2|}_{b_2} - \underbrace{|a_1^* b_1 b_2 a_2^*|}_{a_1} + \underbrace{|a_1^* b_1^* b_2 a_2|}_{a_1} + \underbrace{|a_1^* b_2 a_2 a_2^*|}_{b_2} \\
& + \underbrace{|b_1 b_1^* b_2 b_2^*|}_{a_1} + \underbrace{|b_1 b_2 b_2^* a_2^*|}_{b_2} - \underbrace{|b_1^* b_2 b_2^* a_2|}_{b_2} + \underbrace{|b_2 b_2^* a_2 a_2^*|}_{a_1} \Bigg) \\
& - \left(\underbrace{|a_1 a_1^* b_1 b_1^*|}_{a_1} + \underbrace{|a_1 a_1^* b_1 a_2^*|}_{b_2} - \underbrace{|a_1 a_1^* b_1^* a_2|}_{b_2} - \underbrace{|a_1 a_1^* a_2 a_2^*|}_{a_1} \right. \\
& - \underbrace{|a_1 b_1 b_1^* b_2^*|}_{b_2} - \underbrace{|a_1 b_1 b_2^* a_2^*|}_{a_1} + \underbrace{|a_1 b_1^* b_2^* a_2|}_{a_1} - \underbrace{|a_1 b_2^* a_2 a_2^*|}_{b_2} \\
& + \underbrace{|a_1^* b_1 b_1^* b_2|}_{b_2} + \underbrace{|a_1^* b_1 b_2 a_2^*|}_{a_1} - \underbrace{|a_1^* b_1^* b_2 a_2|}_{a_1} + \underbrace{|a_1^* b_2 a_2 a_2^*|}_{b_2} \\
& - \underbrace{|b_1 b_1^* b_2 b_2^*|}_{a_1} + \underbrace{|b_1 b_2 b_2^* a_2^*|}_{b_2} - \underbrace{|b_1^* b_2 b_2^* a_2|}_{b_2} - \underbrace{|b_2 b_2^* a_2 a_2^*|}_{a_1} \Bigg)
\end{aligned}$$

$$= -8 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| + 8 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|$$

minus combination: i^3a_1 and $e^3a_1 - e^3a_1$ and i^3a_1

$$\begin{aligned} &+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \\ &- \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \end{aligned}$$

[illegible]

$$\begin{aligned}
&= -4 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + 4 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + 4 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + 4 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
&\qquad - 4 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - 4 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| - 4 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + 4 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right|
\end{aligned}$$

plus combination: $i^3 a_1$ and $e^3 b_1 + e^3 b_1$ and $i^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(\begin{array}{c} +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 4 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| - 4 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 4 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\
& \quad - 4 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 4 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| + 4 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 4 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

minus combination: $i^3 a_1$ and $e^3 b_1$ - $e^3 b_1$ and $i^3 a_1$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& - \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& + \left(\begin{array}{c} +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 4 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| + 4 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 4 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| \\
& \quad - 4 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 4 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 4 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 4 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right|
\end{aligned}$$

plus combination: $i^3 a_1$ and $i^3 b_1 + i^3 b_1$ and $i^3 a_1$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

$$+ \left(\begin{array}{c} +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \end{array} \right)$$

$$= +4 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 4 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| - 4 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 4 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right|$$

$$+ 4 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 4 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| - 4 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 4 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|$$

minus combination: $i^3 a_1$ and $i^3 b_1$ - $i^3 b_1$ and $i^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(\begin{array}{c} +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 4 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| + 4 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 4 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| \\
& \quad + 4 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 4 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 4 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 4 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right|
\end{aligned}$$

identical $e^3 a_1$ and $e^3 a_1$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1 a_1^* b_2 b_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| \\
& \quad - 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| \\
& \quad + 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{b_1 b_1^* a_2 a_2^*}_{a_1} \right|
\end{aligned}$$

plus combination: $e^3 a_1$ and $e^3 b_1 + e^3 b_1$ and $e^3 a_1$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

$$+ \left(\begin{array}{c} +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \end{array} \right)$$

$$= +4 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 4 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| + 4 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 4 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right|$$

$$- 4 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 4 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| - 4 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 4 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|$$

minus combination: $e^3 a_1$ and $e^3 b_1$ - $e^3 b_1$ and $e^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(\begin{array}{c} +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ + 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ - 2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ + 2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 4 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 4 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 4 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| \\
& \quad - 4 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 4 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 4 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 4 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right|
\end{aligned}$$

plus combination: $e^3 a_1$ and $i^3 b_1 + i^3 b_1$ and $e^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(\begin{array}{c} +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ +2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ +2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ +2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \\ +2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \\ +2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \\ +2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| + 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\ +2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \end{array} \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| - 4 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| + 4 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 4 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \\
& \quad + 4 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| - 4 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| + 4 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 4 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right|
\end{aligned}$$

minus combination: $e^3 a_1$ and $i^3 b_1$ - $i^3 b_1$ and $e^3 a_1$

$$\begin{aligned}
& + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
& - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
& + \left(+2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \right) \\
& + \left(+2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| - 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \right) \\
& - \left(+2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \right) \\
& - \left(+2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \right) \\
& - \left(-2 \left| \underbrace{a_1 a_1^* b_1 b_2^*}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1^* b_2 b_2^* a_2}_{a_2} \right| \right) \\
& + \left(-2 \left| \underbrace{a_1 a_1^* b_1^* b_2}_{a_2} \right| + 2 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| + 2 \left| \underbrace{a_1 b_2 b_2^* a_2^*}_{a_2} \right| \right) \\
& - \left(-2 \left| \underbrace{a_1^* b_1 b_1^* a_2}_{a_2} \right| - 2 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| + 2 \left| \underbrace{b_1 b_2^* a_2 a_2^*}_{a_2} \right| \right) \\
& + \left(-2 \left| \underbrace{a_1 b_1 b_1^* a_2^*}_{a_2} \right| - 2 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right| + 2 \left| \underbrace{b_1^* b_2 a_2 a_2^*}_{a_2} \right| \right)
\end{aligned}$$

$$\begin{aligned}
& = +4 \left| \underbrace{a_1 a_1^* b_2^* a_2}_{b_1} \right| + 4 \left| \underbrace{a_1^* b_1 b_2 b_2^*}_{b_1} \right| - 4 \left| \underbrace{a_1^* b_1 a_2 a_2^*}_{b_1} \right| - 4 \left| \underbrace{b_1 b_1^* b_2^* a_2}_{b_1} \right| \\
& \quad + 4 \left| \underbrace{a_1 a_1^* b_2 a_2^*}_{b_1} \right| + 4 \left| \underbrace{a_1 b_1^* b_2 b_2^*}_{b_1} \right| - 4 \left| \underbrace{a_1 b_1^* a_2 a_2^*}_{b_1} \right| - 4 \left| \underbrace{b_1 b_1^* b_2 a_2^*}_{b_1} \right|
\end{aligned}$$

identical e^3b_1 and e^3b_1

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| \right) \\
& + \left(\begin{array}{c} + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| + \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| + \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| \\
& + 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| + 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| \\
& + 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|
\end{aligned}$$

plus combination: e^3b_1 and $i^3b_1 + i^3b_1$ and e^3b_1

$$\begin{aligned}
 &+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
 &+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
 \end{aligned}$$

[illegible]

$$= -8 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| + 8 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|$$

minus combination: e^3b_1 and i^3b_1 - i^3b_1 and e^3b_1

$$\begin{aligned} &+ \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\ &- \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \end{aligned}$$

[illegible]

$$\begin{aligned}
&= +4 \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + 4 \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| - 4 \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| + 4 \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\
&\qquad - 4 \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| + 4 \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| - 4 \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - 4 \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right|
\end{aligned}$$

identical $i^3 b_1$ and $i^3 b_1$

$$\begin{aligned}
& + \left(\begin{array}{cc|cc} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{array} \right) - \left(\begin{array}{cc|cc} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{array} \right) + \left(\begin{array}{cc|cc} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| \right) \\
& - \left(\begin{array}{c} - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| - \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| - \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ + \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| - \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| + \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| + \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + \left| \underbrace{a_1 a_1^* b_1 a_2^*}_{b_2} \right| + \left| \underbrace{a_1 a_1^* b_1^* a_2}_{b_2} \right| + \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| \\ - \left| \underbrace{a_1 b_1 b_1^* b_2^*}_{b_2} \right| - \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| - \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| + \left| \underbrace{a_1 b_2^* a_2 a_2^*}_{b_2} \right| \\ - \left| \underbrace{a_1^* b_1 b_1^* b_2}_{b_2} \right| - \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + \left| \underbrace{a_1^* b_2 a_2 a_2^*}_{b_2} \right| \\ + \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| - \left| \underbrace{b_1 b_2 b_2^* a_2^*}_{b_2} \right| - \left| \underbrace{b_1^* b_2 b_2^* a_2}_{b_2} \right| - \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| \end{array} \right) \\
& + \left(-4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right| \right) \\
& = -4 \left| \underbrace{a_1^* b_1 b_2^* a_2}_{a_1} \right| - 2 \left| \underbrace{a_1 a_1^* b_1 b_1^*}_{a_1} \right| + 2 \left| \underbrace{a_1 a_1^* a_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1 b_1 b_2^* a_2^*}_{a_1} \right| \\
& \quad - 2 \left| \underbrace{a_1 b_1^* b_2^* a_2}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1 b_2 a_2^*}_{a_1} \right| - 2 \left| \underbrace{a_1^* b_1^* b_2 a_2}_{a_1} \right| + 2 \left| \underbrace{b_1 b_1^* b_2 b_2^*}_{a_1} \right| \\
& \quad - 2 \left| \underbrace{b_2 b_2^* a_2 a_2^*}_{a_1} \right| - 4 \left| \underbrace{a_1 b_1^* b_2 a_2^*}_{a_1} \right|
\end{aligned}$$